Towards Urdu Full-Duplex Spoken Dialogue: Pipeline, Pilot Fine-Tune, and a Data-Scale Diagnosis

Sheikh Abdul Basit

August 2026

Abstract

No full-duplex (simultaneous listen-and-speak) conversational voice model exists for Urdu, and no channel-separated Urdu conversation dataset exists to train one. This report documents a proof-of-concept that built both: a 32,000-token Urdu SentencePiece tokenizer; a synthetic stereo-dialogue corpus (1,367 two-speaker dialogues, 38.33 hours) assembled by pairing Mozilla Common Voice Urdu clips; and a full fine-tune of Kyutai's Moshi speech-text model, extended to a dual-stream, 16-codebook full-duplex configuration. Training ran for five epochs (110 steps, approximately six GPU-hours, approximately US$17–22) and reached a final loss of 5.39 (text 3.72, audio 1.67). The model emits valid Urdu vocabulary but no coherent sentence structure, consistent with using approximately 0.02–0.13% of the data and compute of the nearest published precedent, J-Moshi. Separately, Kyutai's published evaluation establishes that Moshi itself has measurable gaps from natural human speech in knowledge retention, voice quality, recognition accuracy, safety, and quantized output quality. The analysis therefore distinguishes limitations inherited from the base model from those attributable to this pilot's data and compute constraints.

1. Introduction

1.1 Motivation

The project addresses four documented gaps: the absence of a full-duplex conversational system for Urdu; the lack of a channel-separated Urdu conversation dataset; the lack of a production-quality open-source Urdu text-to-speech system; and the absence of a language-model tokenizer designed for conversational Urdu. The present study focuses on adapting a full-duplex dialogue model and on the dataset limitations governing its performance.

Conventional half-duplex assistants sequence automatic speech recognition, a language model, and text-to-speech, introducing latency and rule-based turn-taking. Moshi instead predicts text, generated audio, and incoming user audio jointly at 80 ms intervals, with turn-taking learned from data (Défossez et al., 2024). Accordingly, this adaptation requires two-channel dialogue training examples rather than ordinary single-channel speech.

1.2 Scope and evidence

The report draws on the project's training log (training2.log; 110 logged optimization steps), decoded generation samples, dataset manifests, build logs, repository documentation, and the cited research literature. Reported project measurements are taken from these primary artifacts. The study is a pilot fine-tune and is not presented as evidence of a deployable Urdu dialogue system.

2. Background and Related Work

2.1 Moshi architecture

Moshi is a speech-text foundation model developed by Kyutai for real-time full-duplex dialogue (Défossez et al., 2024). Its Helium backbone is a seven-billion-parameter language model pretrained on 2.1 trillion English-language tokens. Mimi encodes waveform audio as discrete residual-vector-quantized codes at 12.5 Hz; its first codebook incorporates semantic representations. Moshi's "Inner Monologue" predicts a time-aligned text token ahead of audio tokens within each frame, while a parallel user-audio stream permits simultaneous listening and generation. The source paper reports approximately 160 ms theoretical and 200 ms practical latency.

2.2 Limitations documented for the base model

Kyutai's published benchmarks identify important trade-offs independent of Urdu adaptation (Défossez et al., 2024):

- Knowledge: MMLU accuracy falls from 54.3 for text-only Helium to 49.7 for Moshi (Table 7).
- Voice conditioning: compared with voice-actor conditioning, the public synthetic-voice configuration scores lower on sWUGGY (60.9 vs. 63.0), sBLIMP (54.6 vs. 55.2), spoken StoryCloze (60.9 vs. 62.7), and MMLU (48.7 vs. 49.7).
- Streaming accuracy: reported TTS word-error rate is 4.7% on LibriSpeech test-clean, versus 1.81% for NaturalSpeech 3; streaming ASR reaches 5.7% WER versus 3.6% for the cited Streaming FastConformer baseline (§5.7).
- Speaker identity: 1.3% of model-generated speech segments in a 100-hour evaluation are closer to the user voice than to Moshi's reference voice (§6.3).
- Safety: ALERT aggregate is 83.05, versus 96.95 for GPT-3.5, 99.18 for GPT-4, and 99.98 for Llama 2 (§6.1).
- Quantization: at 2-bit weights, an entropy-based detector marks 83.1% of sampled generations as containing gibberish-audio artifacts (Appendix D).

These findings establish limitations in the pretrained system itself. They should not be attributed to the Urdu fine-tune, nor interpreted as a direct evaluation of the Urdu model.

2.3 Related studies

Study	Relevance
J-Moshi (Ohashi et al., 2025)	Japanese adaptation using a compatible Moshi fine-tuning fork; comparator for training scale and text–audio decoupling.
PersonaPlex (Roy et al., 2026)	Moshi fine-tuning using synthetic and real two-channel data; relevant for interruption-aware audio preparation.
Kyutai DSM (2025)	Delayed-streams modeling for streaming TTS; possible architecture for synthetic Urdu dialogue generation.
Full-Duplex-Bench (Lin et al., 2025)	Evaluation methods for turn-taking, backchannels, and interruption latency.
CharDuplex (Wu et al., 2026)	Character-consistent full-duplex dialogue based on GLM-4-Voice.

3. System and Data-Preparation Pipeline

Common Voice contains isolated read utterances rather than real conversations. To construct training examples, utterances from different speakers were interleaved into two-channel audio, with the agent on the left channel and the user on the right. The resulting pipeline is summarized below.

Stage	Process / output
0	Text corpus: Dolly-15k-ur and pre-existing ur.txt (6.1 GB).
1	Train a 32,000-token Urdu SentencePiece BPE tokenizer.
2	Filter Common Voice v25 Urdu and exclude development/test speakers.
3	Build 57.14 h mono-stitched segments (not used downstream).
4	Create 1,367 stereo dialogues totaling 38.33 h.
5	Create word timestamps using WhisperX large-v3 with Urdu specified.
6	Encode audio with frozen Mimi, eight codebooks at 12.5 Hz.
7	Align Urdu text tokens to word timestamps.
8	Merge arrays into 1,362 parquet training rows; five dialogues dropped.
9	Reinitialize Urdu text embeddings and add a user-audio stream (dep_q=16).
10	Full fine-tuning with DeepSpeed ZeRO-3 for five epochs, checkpoint step_110.
11	Generate and decode ten audio/text continuations.

The dialogue constructor used gender-aware pairing across 40 speaker pairs (14 mixed-gender and 26 same-gender). Model initialization expanded the depth transformer from eight to 16 codebooks: eight for generated audio and eight for the user stream. The selected moshi-finetune fork supports custom tokenizer replacement and text-embedding reinitialization, unlike the cited official repository; additional local patches were applied.

4. Data

Dataset	Hours	Examples / lines	Use
Common Voice v25 Urdu (raw)	294.65	250,779	Source corpus
Filtered manifest	53.33	51,279	Training candidates; dev/test speakers excluded
Mono-stitched segments	57.14	2,264	Not used in final training
Stereo dialogues	38.33	1,367	Synthetic two-speaker audio
Final parquet	—	1,362	Fine-tuning examples
Urdu TTS corpus	9.8	5,311	Acquired but unused
Urdu text corpus	—	~26.65 million lines	Tokenizer training only

The final derived dataset has no held-out validation or test split. Quality was assessed through generated continuations, not a held-out metric. In addition, synthetic turns are nonoverlapping and semantically unrelated: the inactive speaker channel is silent, and interleaved read sentences do not constitute genuine dialogue. Thus, the training data lacks backchannels, interruptions, and coherent conversational context.

5. Experimental Setup

Full-parameter fine-tuning (without LoRA) used DeepSpeed ZeRO-3, fp16 compute and float32 master weights across two GPUs on RunPod A100/H100 instances.

Parameter	Setting
Learning rate	3×10^{-5}; linear warm-up over 100 steps
Batch configuration	2 per device × 16 accumulation × 2 GPUs = effective 64
Training	5 epochs; 110 optimizer steps
Sequence length	128–2,048 frames (maximum ~164 s at 12.5 Hz)
Optimizer	AdamW; $\beta = (0.9, 0.95)$, $\varepsilon = 10^{-5}$, weight decay 0.1
Other settings	Activation checkpointing; text padding 0.5×, semantic codebook 100×, acoustic 1×

6. Results

6.1 Training loss

Total loss declined from 12.99 at optimization step 1 to 5.39 at step 110. At the final recorded step, the text component was 3.72 and the audio component 1.67 (training2.log). Loss reduction alone does not establish dialogue quality, particularly in the absence of held-out evaluation.

Metric	Observed value
Initial total loss	12.99 (step 1)
Final total loss	5.39 (step 110)
Final text loss	3.72
Final audio loss	1.67

6.2 Generated output

Ten approximately 20-second continuations were generated and decoded from the final checkpoint. All ten contained recognizable individual Urdu words but did not exhibit stable sentence-level grammar. Raw decoding also included unresolved-token placeholders. Recurring vocabulary in the samples included words for Punjab, happiness, life, Gwadar, and motorcycle. These qualitative observations indicate that the tokenization and adaptation path produced Urdu lexical material, but they do not demonstrate intelligible full-duplex conversation.

6.3 Comparison with J-Moshi

Measure	J-Moshi (Japanese)	Urdu pilot	Urdu / J-Moshi
Training audio	69,000 h	38.33 h	0.056%
Compute	~4,600 GPU-h	~6 GPU-h	0.13%
Output	Coherent Japanese dialogue	Recognizable words without syntax	Not comparable as metric

The pilot therefore operated far below the nearest cited non-English adaptation in both audio hours and computation. Reported expenditure for the experiment, including preprocessing, was approximately

US$17–22. The sample outputs support successful generation of some Urdu vocabulary; they do not establish that scaling alone will yield functional full-duplex behavior.

7. Discussion

Moshi predicts frame-level text ahead of conditioned audio. When text predictions remain uncertain early in adaptation, the audio path may learn to ignore text conditioning, resulting in text–audio decoupling. The J-Moshi authors describe recovery as text loss decreases to around 3.0; the Urdu pilot ended at 3.72. Two design factors intensify the problem: approximately 95% of frames provide padding or silence rather than direct text supervision, and the 32,000 Urdu text embeddings were initialized anew rather than transferred from a trained Urdu representation.

Two further concerns appear in the project's research materials but were not tested experimentally: Mimi's WavLM-derived semantic representation may inadequately distinguish Urdu phonetic contrasts; and the synthetic corpus does not systematically represent Urdu honorific registers or Urdu–English code-switching. These are hypotheses and data-coverage risks, not measured failure modes of this checkpoint.

8. Limitations and Threats to Validity

- No held-out evaluation: output quality was assessed qualitatively from ten generated samples.
- Checkpoint discrepancy: a retained 3.97 GB fp16 checkpoint and a 16 GB private Hugging Face upload were not reconciled.
- Compute-record inconsistency: the README describes an H100 run of ~5.3 hours and ~105 steps, while training2.log reports two GPUs and 110 steps; logged steps are used here.
- Data loss: five of 1,367 stereo dialogues (about 0.4%) were omitted during parquet assembly by a filename-matching guard, with no detailed drop audit.
- Unused preprocessing: 57.14 hours of mono-stitched audio were generated but not used by the stereo builder.
- Unmeasured conversational behavior: no Full-Duplex-Bench-style turn-taking, backchannel, or interruption assessment was conducted.
- Synthetic dialogue validity: the paired read-speech utterances lack semantic turn-to-turn continuity and overlap.

9. Directions for Future Work

Priority should be given to resolving evaluation and data-quality issues before interpreting further fine-tuning as evidence of dialogue capability. Recommended work includes establishing held-out evaluation data; auditing dropped records and checkpoint consistency; collecting genuine channel-separated conversational Urdu with backchannels and interruptions; assessing Urdu phoneme coverage and social register; evaluating text–audio alignment; and conducting structured listening and full-duplex behavioral tests. Controlled optimization experiments could then assess increased training, adjusted text-loss weighting, and pretrained Urdu embedding initialization. These are research recommendations rather than experimentally validated improvements.

10. Conclusion

This study established a reproducible proof-of-concept for Urdu adaptation of a full-duplex speech-text model: training a dedicated tokenizer, constructing a synthetic two-channel corpus, extending Moshi to a

dual audio stream, and completing a full fine-tune that generated recognizable Urdu words. However, the output did not form coherent dialogue, and the pilot lacked both natural conversational training signals and held-out behavioral evaluation. The result must therefore be interpreted as a pipeline demonstration rather than a working Urdu full-duplex assistant. The strongest next research priority is to obtain substantial authentic two-channel Urdu conversation data, coupled with rigorous measurement of dialogue quality and full-duplex interaction.

References

Défossez, A., Mazaré, L., Orsini, M., Royer, A., Pérez, P., Jégou, H., Grave, E., & Zeghidour, N. (2024). Moshi: A speech-text foundation model for real-time dialogue. arXiv:2410.00037.

Kyutai. (2025). Streaming sequence-to-sequence learning with delayed streams modeling. arXiv:2509.08753.

Lin, et al. (2025). Full-Duplex-Bench: A benchmark to evaluate full-duplex spoken dialogue models on turn-taking. arXiv:2503.04721.

Mozilla. (2026). Common Voice 25.0 (Urdu subset). https://commonvoice.mozilla.org/

Ohashi, A., et al. (2025). Towards a Japanese full-duplex spoken dialogue system (J-Moshi). arXiv:2506.02979.

Roy, et al. (2026). PersonaPlex. NVIDIA. ICASSP 2026.

Wu, D., Liu, Y., Chen, C., Liu, H., & Chng, E. S. (2026). CharDuplex: Building character-consistent full-duplex spoken dialogue models. arXiv:2609.34461.